

Software Engineering - Waterfall model

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Introduction

The Waterfall Model is one of the earliest and most traditional software development life cycle (SDLC) models. It follows a **linear and sequential approach**, where each phase of development must be completed before the next phase begins. Because of its structured nature, the Waterfall Model is easy to understand and manage, making it suitable for projects with well-defined requirements.

Phases of the Waterfall Model

The Waterfall Model is divided into the following phases:

1. **Requirement Analysis**

In this phase, all system requirements are gathered from stakeholders and documented clearly. No design or coding starts until requirements are fully understood.

2. **System Design**

Based on the requirements, the overall system architecture and design specifications are created, including hardware, software, and data flow designs.

3. **Implementation (Coding)**

The system is developed according to the design documents. Each module is coded and tested individually.

4. **Integration and Testing**

All modules are integrated into a complete system and tested to identify defects or errors. This ensures the system meets the specified requirements.

5. **Deployment**

After successful testing, the software is deployed in the real-world environment for use.

by end users.

6. **Maintenance**

Ongoing support is provided to fix bugs, improve performance, or make minor updates after deployment.

Advantages of the Waterfall Model

- Simple and easy to understand
- Well-structured with clearly defined stages
- Easy to manage due to clear milestones
- Suitable for small projects with stable requirements
- Documentation is thorough at every stage

Disadvantages of the Waterfall Model

- Not flexible to changes once development starts
 - Difficult to accommodate changing requirements
 - Testing is done late in the process
 - High risk for complex or long-term projects
 - Not suitable for projects with unclear or evolving requirements
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Types of Projects Suitable for the Waterfall Model

The Waterfall Model is most suitable for:

- Small to medium-sized projects
 - Projects with fixed and well-understood requirements
 - Systems with minimal expected changes
 - Government, healthcare, and banking systems
 - Projects requiring extensive documentation
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Case Study: Hospital Report Management System

Overview

A Hospital Report Management System is used to manage patient records, medical test reports, billing information, and doctor notes. Since hospitals require accuracy, security, and compliance with regulations, the Waterfall Model can be an effective choice when requirements are stable.

Use of Waterfall Model in the Hospital System

- 1. Requirement Analysis**
Hospital administrators, doctors, and lab staff define requirements such as patient registration, report generation, secure data storage, and access control.
- 2. System Design**
Designers create system architecture including databases for patient records, user interfaces for staff, and security mechanisms to protect sensitive data.
- 3. Implementation**
Developers code modules like patient registration, lab reports, billing, and report viewing based on the approved design.
- 4. Integration and Testing**
All modules are integrated and tested to ensure accurate report generation, data

consistency, and compliance with healthcare standards.

5. **Deployment**

The system is installed in the hospital's IT environment and made accessible to authorized users.

6. **Maintenance**

The system is maintained to fix issues, update report formats, and comply with new healthcare regulations.

Conclusion

The Waterfall Model provides a disciplined and systematic approach to software development. While it lacks flexibility, it is highly effective for projects with clear requirements and minimal changes. In the case of a Hospital Report Management System, where accuracy, documentation, and reliability are critical, the Waterfall Model can be successfully applied.